MAT2040
Tutorial 10

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July 8&9, 2024

Question 1

Given

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

(a) compute the determinant of A. Is A nonsingular?

(b) compute adj A and the product A adj$A$.

Solution 1

(a)

$$\det(A) = 1\begin{vmatrix} 3 & 4 \\ 4 & 5 \end{vmatrix} - 2\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} + 3\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 0$$

A is singular.

(b)

$$\operatorname{adj} A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

$$A \operatorname{adj} A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Question 2

Given

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$$

(a) determine the (2, 3) entry of A^{-1} by computing a quotient of two determinants.

(b) compute the third column of A^{-1} by using Cramer's rule to solve $A\mathbf{x} = \mathbf{e_3}$.

Solution 2

(a) Since $A^{-1} = \frac{1}{\det(A)} \operatorname{adj} A$,

$$(A^{-1})_{23} = \frac{1}{\det(A)} C_{32}$$

$$\det(A) = 1 \begin{vmatrix} 4 & 3 \\ 2 & 2 \end{vmatrix} - 0 \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} = 4$$

$$C_{32} = - \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = -3, \qquad (A^{-1})_{23} = -\frac{3}{4}$$

Solution 2

(b)

$$B_1 = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}, \qquad x_1 = \frac{\det(B_1)}{\det(A)} = \frac{2}{4} = \frac{1}{2}$$

$$B_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}, \qquad x_2 = \frac{\det(B_2)}{\det(A)} = -\frac{3}{4}$$

$$B_3 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 4 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \qquad x_3 = \frac{\det(B_3)}{\det(A)} = \frac{4}{4} = 1$$

Question 3

Let B_j denote the matrix obtained by replacing the jth column of the identity matrix with a vector $\mathbf{b} = (b_1, \ldots, b_n)^T$. Use Cramer's rule to show that

$$b_j = \det(B_j) \quad \text{for } j = 1, \ldots, n$$

Solution 3

Consider $I\mathbf{x} = \mathbf{b}$, the solution of it is $\mathbf{x} = \mathbf{b}$.
Given I is nonsingular, it follows from the Cramer's rule that

$$b_j = x_j = \frac{\det(B_j)}{\det(I)} = \det(B_j)$$

Question 4

(a) Let A be a nonsingular $n \times n$ matrix with $n > 1$. Show that

$$\det(\operatorname{adj} A) = (\det(A))^{n-1}$$

Let B be a 3 x 3 matrix. If

$$\operatorname{adj} B = \begin{bmatrix} 1 & 3 & -4 \\ -2 & 2 & -2 \\ 1 & -3 & 4 \end{bmatrix}$$

(b) calculate the value of $\det(\operatorname{adj} B)$. What should the value of $\det(B)$ be? [*Hint*: Use the result from previous question.]

(c) find B.

Solution 4

(a) If $\det(A) = \alpha$, then $\det(A^{-1}) = 1/\alpha$.
Since $A(\operatorname{adj} A) = \alpha$, we have $\operatorname{adj} A = \alpha A^{-1}$. Hence

$$\det(\operatorname{adj} A) = \det(\alpha A^{-1}) = \alpha^n \det(A^{-1}) = \alpha^{n-1} = \det(A)^{n-1}$$

(b)

$$\det(\operatorname{adj} B) = 1\begin{vmatrix} 2 & -2 \\ -3 & 4 \end{vmatrix} - 3\begin{vmatrix} -2 & -2 \\ 1 & 4 \end{vmatrix} - 4\begin{vmatrix} -2 & 2 \\ 1 & -3 \end{vmatrix} = 4.$$

Given $\det(\operatorname{adj} B) = (\det(B))^2$, $\det(B) = \pm 2$.

Solution 4

(c) Since $B(\operatorname{adj} B) = 2$, we have

$$B = \pm 2\,(\operatorname{adj} B)^{-1} = \frac{\pm 2}{\det(\operatorname{adj} B)} \operatorname{adj}(\operatorname{adj} B)) = \pm\frac{1}{2}\operatorname{adj}(\operatorname{adj} B)$$

$$\operatorname{adj}(\operatorname{adj} B) = \begin{bmatrix} 2 & 0 & 2 \\ 6 & 8 & 10 \\ 4 & 6 & 8 \end{bmatrix}$$

$$B = \pm \begin{bmatrix} 1 & 0 & 1 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{bmatrix}$$